Maths Exam Language Bank - Write It Like the Mark Scheme (9MA0)

Oct 15, 2026 · @ET

Purpose: In Maths, getting the right idea is not enough. Marks are awarded for the method shown, for each correct step, and for a clear final statement. This bank shows, for each common situation, what an answer that loses marks looks like next to a full-marks write-up, so Jody can learn to set out her working the way the mark scheme expects.

用法簡述： 每個情境用三欄對照：情境、「常見寫法」（失分）、「完整寫法」（得分）。先蓋住右欄自己寫一次，再對照有沒有漏掉步驟或結論。

How marks are awarded in Maths

Mark	Meaning	What it means for writing
M	Method mark: for using a correct method, even if the arithmetic is wrong	Show the method line, for example the formula and the substitution
A	Accuracy mark: for a correct answer, and it depends on the method mark before it	A wrong method earlier means no accuracy marks later
B	Independent mark: for a fact, value or statement	State the result clearly; it does not depend on a method
dM	Method mark that depends on an earlier method mark	Each stage must be correct for the next to count
ft	Follow through: later marks are allowed after an earlier error	Show every line so the examiner can follow your own working

- **A bare answer can score zero.** When the question says "show that" or "find" with several marks, the working is what is marked.
- **Each line must be readable.** If the examiner cannot see where a number came from, the mark is lost.
- **Say what you have found.** The last line should answer the question, with units and in context where relevant.
- **Use the form asked for.** Exact values, a given form such as ax + by + c = 0, or a rounded value to 3 significant figures: a correct answer in the wrong form can lose the final mark.

How to use this bank

1. Read the situation. Cover the right-hand column and write your own full solution on paper.
2. Uncover the full-marks write-up. Tick each line you wrote and mark any line or sentence that was missing.
3. Close the page and redo the question a day later, writing every line the mark scheme would credit.

The "typical answer" column shows common short answers that lose marks. After the mock paper is marked, replace them with Jody's own shortcuts.

Command words: what each one asks for

Command word	What to write	Mark logic
Find, Calculate, Determine	A method with working, then the answer	Method marks and an accuracy mark
Solve	All solutions in the range, each shown with its working, and checked if needed	A mark for each correct solution; missing solutions lose marks
Show that	Every step from the given information to the stated result, ending with the stated result	The answer is given, so the marks are for the working
Prove	A logical argument from known facts with a conclusion	Marks for the correct structure and a final conclusion
Hence	Use the previous result; an unrelated method may lose the marks	Marks only if the earlier result is used
Hence or otherwise	Use the earlier result or any valid method	Either route earns marks
Exact	Leave surds, fractions, pi and logarithms unevaluated	A decimal answer loses the accuracy mark
Sketch	The shape, with the key points labelled: intercepts, turning points, asymptotes	Marks for the shape and for each labelled feature
State, Write down	A short answer, no working needed	1 mark for the correct value

Command word	What to write	Mark logic
Deduce	A conclusion that follows from earlier results	Marks for a valid conclusion
Explain, Justify, Interpret	A short sentence in context	Marks for the reason, stated in the language of the question

Habits that lose marks, and the fix

Habit	What it costs	Fix
Writing only the answer	Method marks, and every mark if the answer is wrong	Write the formula, the substitution and the result on separate lines
Jumping several steps in one line	Marks for steps the examiner cannot see	Write one step per line, especially in algebra and in "show that"
Misusing the equals sign (for example 2x + 3 = 5 = x)	Accuracy marks, and sometimes the whole method	Each line should equal the line before it; start a new line for a new equation, or use "so"
Rounding early	The accuracy mark on the final answer	Keep full calculator values and round only at the end
Stopping at an intermediate result	The final mark	Re-read the question and answer exactly what it asks, with units
Giving a decimal when an exact answer is asked	The accuracy mark	Leave answers as surds, fractions, pi or ln
Missing solutions in a range	A mark for each missing solution	Find all solutions in the range, and write the range first
Not checking solutions	A mark where a solution is invalid	Substitute back for modulus, logarithm and squared equations, and reject extraneous roots
Calculator in the wrong mode	Every trigonometric answer in the question	Check degrees or radians at the start of each question

Habit	What it costs	Fix
No conclusion or no context	The final mark in proofs, tests and modelling	End with a sentence that answers the question in the context given
Working backwards in "show that"	The method marks	Start from the given information and finish at the stated result
Dropping notation (dx, brackets, limits, +c)	Accuracy marks	Write dx in integrals, brackets around limits, and +c for indefinite integrals

Write-up frames to reuse

Frame	Pattern
Method line	State the formula, then substitute the values, then write the result: for example, the mean = sum of x divided by n = 125 divided by 9 = 13.9
Show that	Start with the given information. Write each algebraic step on its own line. Finish with the stated result and the words "as required".
Conclusion in context	Therefore [the quantity] is [value and units], which means [the answer to the question in words].
Choosing between answers	Reject [value] because [reason from the question]; the solution is [value].
Hypothesis test	H0 and H1 stated, the model under H0, the probability, the comparison, the decision (reject or fail to reject), and the conclusion in context.
Proof by contradiction	Assume the opposite. Show that this leads to a contradiction. Therefore the original statement is true.

Pure Mathematics, Book 1

Situation	Typical answer (loses marks)	Full-marks write-up
Quadratic inequality, $x^2 - 3x - 10 < 0$	$x < 5$ and $x < -2$.	Critical values: $(x - 5)(x + 2) = 0$, so $x = 5$ and $x = -2$. The curve is U-shaped, so it is below zero between the roots: $-2 < x < 5$.
Discriminant	The equation has roots.	$b^2 - 4ac > 0$: two distinct real roots. $b^2 - 4ac = 0$: one repeated root. $b^2 - 4ac < 0$: no real roots. To show real roots exist, show $b^2 - 4ac \geqslant 0$.
Simultaneous equations (line and curve)	$x = 2$, $y = 5$.	Substitute the linear equation into the other to give a quadratic in one variable. Solve it, then substitute each value back to find both coordinates. Give each solution as a pair: (x, y).
Circle: centre and radius	Centre $(-3, 2)$, radius 25.	Complete the square on x and on y: $x^2 + y^2 - 6x + 4y - 12 = 0$ gives $(x - 3)^2 + (y + 2)^2 = 25$. The centre is $(3, -2)$ and the radius is $\sqrt{25} = 5$. The signs inside the brackets are reversed, and the radius is the square root.
Tangent to a circle	Use the gradient of the circle.	Find the gradient of the radius to the point. The tangent is perpendicular to the radius, so its gradient is the negative reciprocal. Use $y - y1 = m(x - x1)$ and rearrange to the form asked.
Differentiation from first principles	Differentiate normally.	$f'(x)$ = the limit as $h \to 0$ of $[f(x + h) - f(x)] / h$. Expand $f(x + h)$, simplify the numerator, divide by h, then let $h \to 0$, writing "as $h \to 0$" before the limit.

Situation	Typical answer (loses marks)	Full-marks write-up
Stationary points	Find $f'(x) = 0$.	Differentiate, set $f'(x) = 0$ and solve for x. Substitute each x into f(x) to find the y-coordinate, and give each point as coordinates.
Nature of stationary points	It is a maximum.	Find $f''(x)$. $f''(1) = -6 < 0$, so (1, 6) is a maximum. $f''(2) = 6 > 0$, so (2, 5) is a minimum. If $f'' = 0$, check the sign of the gradient either side of the point.
Tangent and normal	The normal has the same gradient.	At x = a find the gradient $m = f'(a)$. The tangent has gradient m. The normal has gradient $-1/m$. Use $y - y1 = m(x - x1)$ with the point on the curve.
Increasing function	The graph goes up.	A function is increasing where $f'(x) > 0$. Solve the inequality $f'(x) > 0$ and state the range of x.
Integration with a boundary condition	$y = 2x^3 - 2x^2 + 3x$.	Integrate each term (increase the power by 1 and divide by the new power) and add +c. Substitute the given point to find c, then write the full equation of y.
Definite integral and area	The area is the integral.	Integrate, write the result in square brackets with the limits, then substitute the upper limit and subtract the lower limit. An area below the x-axis gives a negative value, so take the positive value for the area.
Exponential equation that is a hidden quadratic	3^(2x) − 4(3^x) + 3 = 0, so x = 1 or 3.	Let y = 3^x, so $y^2 - 4y + 3 = 0$. Then $(y - 1)(y - 3) = 0$, so y = 1 or 3. Substitute back: 3^x = 1 gives x = 0 and 3^x = 3 gives x = 1.

Situation	Typical answer (loses marks)	Full-marks write-up
Logarithms	Divide both sides by a.	Take logarithms of both sides. For a^x = b, x log a = log b, so x = log b / log a. Use log a + log b = log(ab) and n log a = log(a^n).
Transformations	f(x + 2) moves the graph right.	f(x + 2) is a translation of 2 units to the left. 2f(x) is a stretch with scale factor 2 parallel to the y-axis. f(2x) is a stretch with scale factor 1/2 parallel to the x-axis. −f(x) is a reflection in the x-axis.
Trigonometric equation	sin x = 0.5 gives x = 30°.	Use the graph or CAST diagram to find all solutions in the range: x = 30° and 150°, adding 360° where needed. For sin 2x, change the range first (0 ⩽ 2x ⩽ 720°), solve for 2x, then halve.
Binomial expansion, positive integer n	Multiply out the brackets.	(a + bx)^n = a^n + nC1 a^(n−1)(bx) + nC2 a^(n−2)(bx)² + … Put the whole term bx in brackets so powers and coefficients are applied to both b and x.
Vectors	They are the same length.	The magnitude of a vector (x, y) is √(x² + y²). To show three points are collinear, show that AB is a multiple of BC and that the vectors share the point B.
Sketching a curve	A smooth curve.	Mark the intercepts with both axes, the turning points, any asymptotes, and the behaviour as x → ±∞. Label the coordinates, and show the shape correctly (for example, a cubic with a positive x³ term goes from bottom left to top right).

Pure Mathematics, Book 2: Chapters 1-4

Chapter names follow the usual Pearson order; check them against Jody's textbook.

Situation	Typical answer (loses marks)	Full-marks write-up
Factor theorem (Ch 1)	f(2) is zero, so it works.	f(2) = 0, so by the factor theorem (x − 2) is a factor of f(x). To find the other factors, divide f(x) by (x − 2) and factorise the quotient.
Proof by contradiction (Ch 1)	Write a quick argument.	Assume the statement is false, so [the opposite is true]. Deduce a consequence by correct steps. State the contradiction. Therefore the original statement is true.
Partial fractions (Ch 1)	Split the top.	Write the form, for example (11x − 1) / ((x + 1)(2x − 1)) = A/(x + 1) + B/(2x − 1). Multiply through by the denominator: 11x − 1 = A(2x − 1) + B(x + 1). Substitute x = −1 and x = 1/2 to find A and B. For a repeated factor use A/(x + 1) + B/(x + 1)². If the numerator has degree at least that of the denominator, divide first.
Domain and range (Ch 2)	The domain is the answer.	The domain is the set of allowed inputs (x values); the range is the set of outputs (f(x) values). State them as inequalities, for example x ⩾ 2 and f(x) ⩾ 0.
Composite function (Ch 2)	fg(x) means f times g.	fg(x) = f(g(x)): apply g first, then f. Substitute the whole expression g(x) into f.

Situation	Typical answer (loses marks)	Full-marks write-up
Inverse function (Ch 2)	Swap the letters.	Write y = f(x), make x the subject, then replace y by x to give $f^{-1}(x)$. The domain of f^{-1} is the range of f. The graph of f^{-1} is the reflection of f in the line y = x.
Modulus equation (Ch 2)	Solve it like a normal equation.	For mod(2x − 5) = x + 1, solve 2x − 5 = x + 1 to give x = 6, and −(2x − 5) = x + 1 to give x = 4/3. Substitute each solution into the original equation to check, because the right-hand side must not be negative.
Graph of a modulus function (Ch 2)	The graph is the same.	For y = mod(f(x)), keep the parts of the graph above the x-axis and reflect the parts below the x-axis in the x-axis. The graph never goes below the x-axis.
Arithmetic sequence (Ch 3)	Use the formula.	The nth term is a + (n − 1)d. The sum of n terms is (n/2)[2a + (n − 1)d]. Write down a and d first, then substitute.
Geometric sequence (Ch 3)	Use the formula.	The nth term is ar^(n − 1). The sum of n terms is a(1 − r^n)/(1 − r). The sum to infinity is a/(1 − r), valid only if mod(r) < 1, which should be stated.
Sigma notation (Ch 3)	The sum of the numbers.	The sum for r = 1 to n of u_r means u_1 + u_2 + ... + u_n. Identify whether the terms are arithmetic or geometric, find the first term and the common difference or ratio, then use the sum formula.

Situation	Typical answer (loses marks)	Full-marks write-up
Recurrence relations (Ch 3)	Guess the pattern.	Substitute each term into the rule to find the next: $u_{n+1} = f(u_n)$. A sequence is increasing if $u_{n+1} > u_n$ for all n, and periodic if the terms repeat after a fixed number of steps.
Modelling with series (Ch 3)	A long calculation.	Identify the type of sequence from the situation (a constant difference is arithmetic, a constant ratio is geometric). State the first term and the common difference or ratio, then use the formula and answer in the context given.
Binomial expansion for any n (Ch 4)	Expand like a positive power.	$(1 + x)^n = 1 + nx + [n(n - 1)/2!]x^2 + [n(n - 1)(n - 2)/3!]x^3 + \ldots$ Put the whole term in brackets when substituting, such as (3x) for x.
Validity of the expansion (Ch 4)	It works for all x.	The expansion of $(1 + ax)^n$ is valid when $\text{mod}(ax) < 1$, which gives $\text{mod}(x) < 1/\text{mod}(a)$. For $(a + bx)^n$ factor out a^n first, giving $a^n(1 + (b/a)x)^n$, valid for $\text{mod}(bx/a) < 1$.
Binomial and partial fractions (Ch 4)	Expand the fraction.	Split into partial fractions first, expand each term separately using the binomial expansion, then add the series term by term. State the range of validity as the intersection of the two ranges.
Approximating with a binomial expansion (Ch 4)	Use any value of x.	Choose a value of x within the range of validity, substitute it into the expansion, and check it gives a value close to the exact value on the calculator.

Pure Mathematics, Book 2: Chapters 5-9

Situation	Typical answer (loses marks)	Full-marks write-up
Arc length and sector area	Writes 12.6 with no working, or uses degrees in the radian formula	State the formula with the angle in radians: arc length s = r theta = 8 x 1.2 = 9.6 cm. Area of sector A = 1/2 r squared theta = 1/2 x 64 x 1.2 = 38.4 cm squared. Always include units.
Area of a segment	Subtracts areas without saying which shapes they are	Area of segment = area of sector minus area of triangle = 1/2 r squared theta minus 1/2 r squared sin theta = 1/2 r squared (theta minus sin theta). Then substitute and give the answer to 3 significant figures with units.
Small angle approximation	Replaces sin x with x everywhere with no comment	For small x in radians, sin x is approximately x, cos x is approximately 1 minus x squared over 2, and tan x is approximately x. Substitute these, simplify, and state that the approximation is valid because x is small and measured in radians.
Equation with sec, cosec or cot	Leaves it in sec form and cannot solve it	Rewrite using sec x = 1 over cos x (or cosec x = 1 over sin x, cot x = cos x over sin x). Use the identity 1 plus tan squared x = sec squared x or 1 plus cot squared x = cosec squared x to form a quadratic in one trig function, factorise, solve, then list all solutions in the given interval.

Situation	Typical answer (loses marks)	Full-marks write-up
Inverse trig functions	Gives an answer outside the range or forgets the range	State the principal range: arcsin x gives values from minus pi/2 to pi/2, arccos x from 0 to pi, arctan x from minus pi/2 to pi/2. Domain of arcsin and arccos is minus 1 to 1. Quote the range when sketching or evaluating.
Prove a trig identity	Works on both sides at once, or moves terms across the equals sign	Start with the left-hand side only. Use a stated identity (for example sin squared x plus cos squared x = 1), simplify line by line, and arrive at the right-hand side. Finish with: = RHS, as required. Never treat the identity as an equation to solve.
Compound angle formulae	Writes sin(A + B) = sin A + sin B	Quote the correct formula: sin(A + B) = sin A cos B + cos A sin B. Substitute the known values, show the exact values used, and simplify to an exact answer when the question asks for it.
Double angle formulae	Chooses a form of cos 2x without reason	State cos 2x = cos squared x minus sin squared x = 2 cos squared x minus 1 = 1 minus 2 sin squared x. Say which form you are using and why, for example: I use 2 cos squared x minus 1 so that the equation is a quadratic in cos x.

Situation	Typical answer (loses marks)	Full-marks write-up
Write a sin x plus b cos x as R sin(x + alpha)	Gives R and alpha with no working	Let a sin x + b cos x = R sin(x + alpha) = R sin x cos alpha + R cos x sin alpha. Compare coefficients: R cos alpha = a and R sin alpha = b. So R = square root of (a squared + b squared) and tan alpha = b over a, with alpha in the correct quadrant (state acute if required).
Maximum or minimum from R sin(x + alpha)	States the maximum value but not the value of x	The maximum value is R, since sin(x + alpha) has maximum 1. This occurs when x + alpha = 90 degrees, so x = 90 degrees minus alpha. The minimum value is minus R when x + alpha = 270 degrees. Give both the value and where it occurs.
Trigonometric modelling (for example tide or temperature)	Gives a number with no meaning	Substitute into the model, solve, and give the answer in context: The depth is a maximum of 7.2 m, which occurs at 14:30. Interpret constants: R is the amplitude, alpha shifts the graph, and the period is 360 over the coefficient of t. State any limitation of the model.
Parametric to Cartesian	Eliminates t but leaves a messy relation	Make t the subject of one equation, substitute into the other, and simplify to give y in terms of x. Or use a trig identity such as cos squared t + sin squared t = 1. State the Cartesian equation and any restriction on x or y from the range of t.

Situation	Typical answer (loses marks)	Full-marks write-up
Parametric gradient and tangent	Differentiates x and y but does not divide	dy/dx = (dy/dt) divided by (dx/dt). Substitute the value of t to find the gradient m, find the point (x, y) at that t, then use y minus y1 = m(x minus x1). The normal has gradient minus 1 over m.
Parametric area or intersection	Uses the wrong limits	Change the limits to t values: when x = a, t = t1, and when x = b, t = t2. Use area = integral of y (dx/dt) dt between t1 and t2. State the limits conversion explicitly.
Chain rule	Differentiates the outside only	Let u = inner function, so y = f(u). dy/dx = (dy/du) x (du/dx). For example, y = (3x + 1) to the power 5: dy/dx = 5(3x + 1) to the power 4 x 3 = 15(3x + 1) to the power 4.
Product rule	Differentiates each factor and multiplies	Let u = ... and v = ..., so u′ = ... and v′ = ... Then dy/dx = u′v + uv′. Substitute, then factorise fully if the question asks for a stationary point.
Quotient rule	Gets the numerator order wrong	For y = u over v, dy/dx = (u′v minus uv′) over v squared. Write u, v, u′ and v′ first, then substitute so the order of subtraction is clear.
Differentiating exponentials and logs	Forgets the chain rule or constant	d/dx of e to the power kx = k e to the power kx. d/dx of a to the power x = a to the power x ln a. d/dx of ln x = 1 over x. d/dx of ln(f(x)) = f′(x) over f(x). Quote the rule used.

Situation	Typical answer (loses marks)	Full-marks write-up
Implicit differentiation	Differentiates y terms as if they were x terms	Differentiate every term with respect to x. Differentiate y squared as 2y dy/dx and xy using the product rule as y + x dy/dx. Collect all dy/dx terms on one side, factorise and make dy/dx the subject.
Rates of change (related rates)	Works out dV/dr and stops	Write what is wanted, for example dr/dt. Use the chain rule: dr/dt = (dr/dV) x (dV/dt), or dV/dt = (dV/dr) x (dr/dt). Substitute the values and give the answer with units: The radius is increasing at 0.05 cm per second.
Convex and concave	Says the graph is curved	f''(x) greater than 0 means f is convex (concave up) on that interval. f''(x) less than 0 means f is concave (concave down). A point of inflection occurs where f'' changes sign. Show the sign change before stating the point is an inflection.
Numerical methods (iteration)	Writes the final value only	Show the change of sign: f(a) and f(b) have opposite signs and f is continuous, so a root lies in the interval (a, b). For iteration, show each value x1, x2, x3 to the required accuracy. To justify accuracy, test a change of sign in an interval one unit in the last decimal place either side.

Statistics and Mechanics (Applied Book 1)

Statistics

Situation	Typical answer (loses marks)	Full-marks write-up
Sampling methods	Says random sample is fair	Name the method and give the reason. Simple random sample: every member has an equal chance of selection, using a numbered list and random numbers. Stratified: population split into groups and sampled in proportion to group size. Opportunity sampling may be biased because it only includes people available at that time.
Describe a sample or population	Mixes up the two words	A population is the whole set of items of interest. A sample is a subset of the population. A census observes every member of the population. A sampling frame is a list of all members of the population.
Quartiles and outliers	Calls any extreme value an outlier	An outlier is a value more than 1.5 x IQR above Q3 or below Q1. Calculate Q1 minus 1.5 x IQR and Q3 plus 1.5 x IQR, then state which values lie outside these limits. (Use the rule the question gives.)
Standard deviation	Says it is the average distance from the mean	Standard deviation measures the spread of the data about the mean. A larger standard deviation means the data are more spread out. For coded data, y = (x minus a)/b gives mean of x = b x mean of y + a and SD of x = b x SD of y.

Situation	Typical answer (loses marks)	Full-marks write-up
Compare two data sets	Says A is bigger than B	Compare a measure of location and a measure of spread, in context: On average, the journey times in town A are longer (median 24 against 19 minutes), and they are more varied (IQR 12 against 7 minutes).
Interpret a histogram	Reads heights as frequency	Use frequency density = frequency divided by class width. Frequency = frequency density x class width = area of the bar. State the scale used and show the calculation for each bar.
Correlation	States positive correlation only	There is a (weak or strong) positive linear correlation between x and y: as the daily temperature increases, ice-cream sales tend to increase. Correlation does not imply causation.
Regression line interpretation	Quotes the equation only	In y = a + bx, b is the change in y for each one-unit increase in x. Say it in context: for each extra hour of revision, the mark increases by 4.2 on average. The intercept a is the predicted y when x = 0 (meaningful only if x = 0 is realistic).
Interpolation and extrapolation	Predicts without comment	Interpolation (x inside the data range) is reliable. Extrapolation (x outside the range) is unreliable because the linear relationship may not continue. State which one you are doing.

Situation	Typical answer (loses marks)	Full-marks write-up
Probability and Venn diagrams	Writes probabilities with no working	Define the events, fill the Venn diagram from the intersection outwards, and check the total is 1. For independence show P(A and B) = P(A) x P(B). For mutually exclusive events P(A and B) = 0.
Conditional probability	Divides by the wrong thing	P(A given B) = P(A and B) divided by P(B). Write the formula first, identify the two probabilities, substitute and simplify to a fraction or decimal.
Binomial model: state the conditions	Says there are two outcomes	A binomial model is suitable if there is a fixed number of trials n, each trial has two outcomes (success or failure), the trials are independent, and the probability of success p is constant. Relate each point to the context.
Binomial probability	Uses the wrong cumulative notation	Define X ~ B(n, p). Then write P(X = 3), or P(X less than or equal to 4), and evaluate. For P(X greater than 4) use 1 minus P(X less than or equal to 4).
Hypothesis test: set up	Writes H0 in words only	Define the parameter in context: Let p be the probability that a seed germinates. H0: p = 0.4, H1: p greater than 0.4 (one-tailed). State the distribution under H0: X ~ B(20, 0.4).

Situation	Typical answer (loses marks)	Full-marks write-up
Hypothesis test: conclusion	Says reject H0 so the claim is true	Compare the p-value with the significance level, then state: Since 0.032 is less than 0.05, reject H0. There is sufficient evidence at the 5 percent level to suggest that the probability of germination has increased. Never say proved.
Critical region	Gives the region without the probability	Find the largest or smallest value with probability not exceeding the significance level. For example P(X greater than or equal to 13) = 0.021 less than 0.05, while P(X greater than or equal to 12) = 0.057 greater than 0.05, so the critical region is X greater than or equal to 13. Also state the actual significance level.
Test value in the critical region or not	Says the result is not significant	Because the observed value 11 is not in the critical region, there is insufficient evidence to reject H0. State the conclusion in context: there is no evidence that the proportion of defective items has changed.

Mechanics

Situation	Typical answer (loses marks)	Full-marks write-up
Modelling assumptions	Says it is a perfect model	Name the assumption and its meaning. Particle: the object has mass but no size, so rotation and air resistance are ignored. Light string: negligible mass so the tension is the same throughout. Smooth surface: no friction. Inextensible string: same acceleration for connected particles.
Using suvat	Chooses a formula without listing values	List s, u, v, a and t with the unknown marked. Choose the equation that has no unwanted variable, for example v squared = u squared + 2as. Substitute and solve, then state direction and units.
Newton's second law	Writes F = ma without resolving	Draw a force diagram. Resolve in the direction of motion: R(up the slope) T minus mg sin 30 = ma. State the direction you are resolving in each time.
Connected particles	Treats the system as one object only	Write an equation of motion for each particle, using the same acceleration magnitude and the same tension (light inextensible string and smooth pulley). Then add or substitute to solve for a and T. Check by substituting back.
Friction	Uses F = mu R without limiting	When the particle is on the point of moving or is moving, F = mu R. Otherwise F is less than or equal to mu R. Find R first by resolving perpendicular to the surface.

Situation	Typical answer (loses marks)	Full-marks write-up
Variable acceleration	Uses suvat	suvat applies only for constant acceleration. For variable acceleration, differentiate: v = dr/dt and a = dv/dt. Integrate to go back: v = integral of a dt + C. Use the initial conditions to find C.
Velocity-time graphs	Reads values without meaning	Gradient = acceleration. Area under the graph = displacement (use signed area) or distance (use total area). Show the area calculation as trapezium areas with the dimensions listed.
Projectile or vertical motion	Ignores the sign convention	State the direction taken as positive, for example: take upwards as positive so a = minus 9.8 m per second squared. Keep the same signs throughout and interpret a negative answer: the particle is moving downwards.
Moments (equilibrium)	Ignores a force	Take moments about a named point: sum of clockwise moments = sum of anticlockwise moments. Include each force with its perpendicular distance and state the model used (uniform rod, weight acts at the midpoint).
Interpret the answer in context	Stops at a number	Return to the question: The particle reaches the ground after 2.3 seconds. Check the answer is realistic, give units and the required accuracy (g = 9.8 unless told otherwise, give 2 or 3 significant figures).

Proof and Communication Skills

Situation	Typical answer (loses marks)	Full-marks write-up
Show that	Works backwards from the answer or jumps to the printed result	Start from the given information, show every step, and arrive at the printed result. State the final line as the given expression. Do not use the printed answer in your working.
Hence	Uses a different method	Use the previous part. The word hence means you must use the result just proved. Quote it: Using the result from part (a), ...
Exact answer	Gives a decimal from a calculator	Leave the answer in surds, fractions, pi or ln form, for example 3 root 2 over 2 or 2 ln 3. Do not round.
Proof by deduction	Tests a few numbers	Start with a general case. Let n be an integer, so an even number is 2n and an odd number is 2n + 1. Expand, simplify, and conclude: this is a multiple of 2 so it is even.
Proof by exhaustion	Lists only some cases	Split into all possible cases (for example n even and n odd). Prove each case separately, then state: all cases have been considered, so the statement is true for all integers.
Disproof by counterexample	Says it is false	Give one specific value that fails and show the calculation: When n = 3, n squared + n + 1 = 13, which is prime, but when n = 4 the value is 21 = 3 x 7, which is not prime, so the statement is false.

Situation	Typical answer (loses marks)	Full-marks write-up
Proof by contradiction	Assumes the result without the negation	Assume the opposite: suppose that root 2 is rational, so root 2 = a/b where a and b are integers with no common factor. Derive a statement that contradicts this, then conclude: this is a contradiction, so the assumption is false and root 2 is irrational.
Check solutions	Accepts all answers from squaring	When an equation has been squared or involves logs or square roots, substitute each solution back and reject those that fail, saying why: x = 4 is rejected because ln(x minus 5) is not defined.
Rounding	Rounds in the middle of a calculation	Keep full calculator values until the last step. Give the final answer to the accuracy requested (3 significant figures by default for non-exact answers) and show an unrounded value first if possible.
Use of the equals sign	Writes a chain of equals that are not equal	Each equals sign must connect equal quantities. Use a new line for each step, or use the word implies. Do not write 3x = 6 = x = 2.
Conclusion in context	Gives x = 5 only	State the answer in the language of the question: The shop must sell 5 items to break even. Include units and the meaning of the result.

Situation	Typical answer (loses marks)	Full-marks write-up
Sketching graphs	Draws a curve with no labels	Label the axes, mark intercepts with coordinates, mark asymptotes with equations, mark turning points, and show end behaviour. Use a ruler for lines and keep the shape correct in each quadrant.
Notation	Mixes dy/dx and y′ or leaves out dx	Write integral signs with dx, include the constant + C for indefinite integrals, and write limits as the number on the integral bar. Use brackets round negative numbers and powers.
Range restrictions	Gives all solutions of a trig equation	Check the given interval (for example 0 to 360 degrees). Find the principal value, then use the symmetry of the graph to list all solutions in range and discard others.
Calculator mode	Gets radians and degrees wrong	Check the mode before each question: radians for calculus and arc length questions, degrees when angles are given in degrees. Write the mode in your working if there is any doubt.
Correct use of the words in the question	Uses everyday words	Use the exact command word. Find: calculate with working. Show: demonstrate fully. Determine: find with justification. Explain: give a reason. Justify: give evidence for the conclusion.

Daily Practice Routine and Checklist

Three-pass routine for every practice question

1. First pass: attempt the question under timed conditions in your own way.
2. Second pass: open the mark scheme and mark yourself line by line. Write M, A or B next to each step you would have scored and circle every mark you lost.
3. Third pass: rewrite the full solution in mark-scheme style, using the write-up frames in this document. Keep a log of the lost-mark habit (for example: no formula quoted, no units, no conclusion in context).

Self-check before handing in

Check	Question to ask
Formula	Did I write the formula before substituting?
Working	Could someone follow every step without the answer?
Accuracy	Is the answer to the right accuracy, with units?
Exact	Did the question ask for an exact value?
Context	Did I answer in the language of the question?
Range	Are all solutions inside the interval, and did I reject invalid ones?
Show that	Did I end on the printed result with no jump?
Sketch	Are intercepts, asymptotes and turning points labelled?
Reasonableness	Is the answer sensible (probability between 0 and 1, length positive)?

Weekly progress tracker

Week	Topic practised	Marks lost to communication	Habit to fix next week
1			
2			
3			

Week	Topic practised	Marks lost to communication	Habit to fix next week
4			

Final note for the tutor

The typical answer column is inferred from common examiner reports, so after Jody sits the Maths mock paper, replace those entries with her real wording. Chapter names follow the usual Pearson order and may differ slightly from the Roedean teaching sequence.